Embedded Software Engineer – Power Electronics, Energy Storage

Redwood Materials · San Francisco, California, United States

Job ID (this archive)	ENERGY-089
Employer	Redwood Materials
Location(s)	San Francisco, California, United States
Career stage (archive tag)	Entry or experienced (Entry / Mid-career)
Location group	Other U.S.
Majors (archive tags)	Electrical Engineering, Computer Engineering, Computer Science
Energy area	Battery storage / software
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/redwoodmaterials/jobs/6003203004?gh_jid=6003203004
Original link (from the course sheet)	https://boards.greenhouse.io/redwoodmaterials/jobs/6003203004?gh_jid=6003203004
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-060

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Redwood Materials

Redwood Materials was founded in 2017 by JB Straubel, co-founder and former CTO of Tesla, on a simple but radical idea: the critical minerals powering the energy transition already exist. They just need to come back. Today Redwood is the largest lithium-ion battery recycler in North America — recovering, refining, and redeploying critical minerals from end-of-life batteries back into the supply chain. On the energy side, Redwood has deployed the largest second-life battery project in the world, delivering power to AI data centers in months rather than years using proprietary energy architecture engineered entirely in-house. The company is profitable, the market it operates in is projected to grow 20 to 50 times over the next decade, and Redwood is the only company with the supply chain, the technology, and the operational scale to lead it.

Embedded Software Engineer – Power Electronics

We are at the precipice of a global energy transition. At Redwood Materials, we are moving beyond battery recycling to build the most advanced, circular energy ecosystem on the planet. Our goal is to move the industry away from static, hardware-heavy infrastructure toward agile, software-controlled energy assets.

While others build "black box" hardware, we are building software-defined power systems. Our Embedded Software team is responsible for the intelligence that drives our power stages, ensuring our energy storage solutions are safe, efficient, and capable of stabilizing the world's most demanding grids.

As an Embedded Software Engineer, you will be the architect of the real-time firmware that drives our power conversion units. You will live at the intersection of silicon and power, writing the low-level code that controls high-speed switching, manages thermal safety, and communicates with the broader site infrastructure. You will be responsible for bringing new products from white-space R&D to global scale, ensuring our code is as resilient as the hardware it controls.

Responsibilities will include:

- Real-Time Control Firmware: Design and implement bare-metal software applications for power electronics, focusing on high-resolution PWM generation, ADC synchronization, and complex timing logic.
- Low-Level Architecture: Develop and optimize drivers for microcontroller peripherals (SPI, CAN, Ethernet) and architect memory-efficient, multi-core processing strategies for real-time targets.
- Hardware-Software Synergy: Collaborate with power electronics controls engineers to translate controls theory into deterministic, high-performance code running on Cortex-R and

Cortex-M architectures.

- Modern Build & Test: Maintain robust software build procedures and scripts. You will write highly testable code and integrate it into a modern CI/CD pipeline to ensure every line of code is validated before it hits the lab.
- System Optimization: Profile and optimize code across multi-core environments, ensuring maximum task efficiency and minimal latency in time-critical control loops.
- Product Scaling: Leverage a proven track record to take embedded systems from initial prototype to mass-market production, maintaining high standards for code quality and reliability.

Desired Qualifications:

- Background: A BS or MS in Electrical Engineering, Computer Engineering, or Computer Science, combined with 2+ years of experience in firmware engineering and a proven track record of shipping new products at scale.
- Embedded Mastery: Expert proficiency in Rust or C with deep experience debugging complex embedded systems using oscilloscopes, logic analyzers, and JTAG/SWD debuggers.
- Silicon Expertise: Extensive experience with ARM Cortex-M and Cortex-R architectures, including low-level system architecture, multi-core memory optimization, and bare-metal task management.
- Power Electronics Literacy: A deep understanding of power-specific software requirements, specifically high-speed ADC sampling, high-resolution PWMs, and hardware timers.
- Safety-Critical Development: A safety-centered mindset with experience implementing functional isolation and ensuring freedom from interference (FFI) in complex, multi-core software environments. Familiarity with UL certification standards for energy storage systems is a significant plus.
- Communication Protocols: Practical experience with digital protocols such as I2C, UART, SPI, CAN/CAN-FD, and Ethernet. Familiarity with reverse-engineering CAN protocols or developing custom evaluation tools is a significant plus.
- Infrastructure & Automation: Knowledge of software build systems and the ability to maintain build scripts (e.g., Make, CMake). Experience with test automation and CI/CD is essential.
- Bonus Qualifications: Experience developing bootloaders, implementing secure boot, or profiling code on real-time, multi-core targets.

In accordance with California pay transparency laws, the salary range for this position is listed below. Actual compensation may vary based on a variety of factors, including experience, education, and skills.

California Pay Range:

$180,000—$237,500 USD

The position is full-time. Compensation will be commensurate with experience.

We collect personal information (PI) from you in connection with your application for employment with Redwood Materials, including the following categories of PI: identifiers, personal records, professional or employment information, and inferences drawn from your PI. We collect your PI for our purposes, including performing services and operations related to your potential employment. If you have additional privacy-related questions, please contact us at privacy@redwoodmaterials.com.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Background: A BS or MS in Electrical Engineering, Computer Engineering, or Computer Science, combined with 2+ years of experience in firmware engineering and a proven track record of shipping new products at scale. | Embedded Mastery: Expert proficiency in Rust or C with deep experience debugging complex embedded systems using oscilloscopes, logic analyzers, and JTAG/SWD debuggers.

Key skills listed in the archive: Rust; R; optimization; automation

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 6003203004

Career stage: Entry / Mid-career

Major fit: Direct skills pathway: software development, computing and engineering; consult the exact degree and experience requirements. Energy-sector connection: Control, integration and infrastructure software for stationary energy storage.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 6003203004.

Sources and retrieval details

Employer posting

https://boards.greenhouse.io/redwoodmaterials/jobs/6003203004?gh_jid=6003203004

Retrieved source

https://boards-api.greenhouse.io/v1/boards/redwoodmaterials/jobs?content=true

Retrieved: 2026-10-26T16:33:09.192514+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.